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In Memory of Bernard Schmitt: A Survey of his Quantum Monetary Approach to Macroeconomics

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ABSTRACT
Bernard Schmitt (born in Colmar on 6 November 1929) died in Beaune, France, on 26 March 2014. This paper is dedicated to his memory and to his ground-breaking contribution to macroeconomic analysis. It provides a summary of Schmitt's main results as well as an introduction to his theory and a comparison with mainstream economics. The paper follows a line ranging from the main arguments developed by Schmitt in his positive analysis of national and international macroeconomics to the reform plans he advocated in his normative (regulatory) analysis. It also considers his critical analysis of general equilibrium and the relationship between his and Keynes's macroeconomic theory. Money, profit, capital, and the anomalies affecting today's capitalism, namely, inflation and unemployment, are some of the topics investigated in Schmitt's analysis of national economic systems. Countries' sovereign debt, together with the payment of interest on countries' foreign debt and the monetary disorders caused by the absence of a properly-working international payment system, are the topics of Schmitt's analysis of international economics addressed in this paper. The further points considered here are Schmitt's proposal for a reform of the national payments system and his proposals for a worldwide and a single-country reform of the international payments system.

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1. Introduction

In a research career that spanned over more than sixty years, Bernard Schmitt developed a monetary macroeconomic analysis that ranged from the study of bank money to the proposal for a one-country monetary reform of cross-border payments, passing through an original investigation of the main tenets of national and international economic systems. He contributed to positive, critical, and normative analyses. His

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quantum macroeconomic approach pertains to the first category and provides an explanation of money, income, value, prices, profit, saving, investment, capital, amortization, and international payments. The analyses of relative prices and of the neoclassical interpretation of Keynes's thought pertain to the second category, whereas the analyses of inflation, unemployment, and countries' external and sovereign debt pertain to both the first and the second category. Finally, his proposals for a national and an international monetary reform are the main features of the third category. In this paper, we honour Schmitt's memory by summarizing most of his crucial contributions to the building of quantum macroeconomics (see also Bailly, Cencini, and Rossi 2017).

2. The Theory of the Monetary Circuit

The analysis of bank money is Schmitt's starting point and the pivotal element of his entire research work in macroeconomics. Reassessing the classical distinction between nominal and real money, he comes to define money as a numerical form deprived of any intrinsic value and issued by banks as an asset–liability. It is double-entry accounting that enables banks to issue money as their spontaneous acknowledgment of debt. The metaphysical miracle of creating a positive value out of nothing is avoided by the simultaneous emission of positive and negative numbers entered by banks on the assets and liabilities' side of their balance sheets. For the emission of money to give rise to a positive outcome, money must be associated with production. It is only if the numerical form acquires a real content that nominal money is transformed into real money, that is, income.

Schmitt shows that it is through the payment of wages that money and physical output become the terms of an absolute exchange, where production and money define one another. The result of production is a sum of money: production is an instantaneous event through which the product acquires a numerical expression and becomes the very object of monetary wages.

Money is created and cleared in any payment, since its nature is that of a flow. The dated idea that money is a stock put into circulation is discarded by Schmitt for the simple reason that money is a valueless means of payment, the presence of which is required only to convey payments, not to finance them. What can be stocked is real money, that is to say, income. The object of bank deposits is income and not money as such, and income is nothing other than the 'product-into-the-money'.

Contrary to what the Classics believed, Schmitt shows that economic value is not a substance, a mysterious dimension goods and services would acquire through their production. Economic value is the numerical expression of products, while money is what is used to introduce numbers in the economy. In themselves, numbers are non-dimensional. Hence, money is a dimensionless numerical form necessary to express the value of produced output. However, it would be wrong to consider money as the standard or unit of measure of economic value. Isolated from production, money would be unfit to express the numerical value of any economic object. Following Keynes (1936), Schmitt identifies the wage-unit as the specific unit of measure of economics. Wage-units are expressed in money terms, of course, but it is because money and output become the terms of an absolute exchange that wages become the measure of produced output, their very economic definition.

These tenets about money became the foundations of the so-called theory of the monetary circuit with roots in Dijon, France, of which Schmitt was the founding father (his first publication, *Le pouvoir d'achat*, dates back to 1960).

The notion of the circuit may be traced back to the Physiocrats (the 18th century French philosophers) and their depiction of the agricultural cycle of production. In contemporary economics, the notion of the circuit denotes firms' investment of funds into production processes and the correlated refund and profit formation through the sale of produced goods and services. In this form, it underpinned many approaches to economics from Marx to Keynes (see Gnos and Rochon 2003).

In the 1970s, some writers, among whom Parguez (1975) and Barrère (1990) in France and Graziani (1989) in Italy, promoted this view to the rank of a fully-fledged theory. According to these authors, the circuit is basically the circuit of monetary income as formed in the payment of workers' wage-bills. Money would be created by banks and lent to firms, the former granting credit to the latter. Money would then be cleared when firms sell the goods produced, including profit-goods, and pay back their loans. Successively created and cleared in banks' books, money is supposed to flow in a circuit, which may be disrupted when the goods produced are in no way demanded in markets or when banks create money in excess.

For his part, Schmitt (1975) had already observed that the representation by which money would be created, circulated, and eventually cleared by banks is inconsistent with reality. Double-entry accounting enforces another story: it entails two instantaneous complementary circuits of money, nominal and real.

Double-entry accounting is mandatory. In any payment, the flow and the backflow of money are simultaneous events. In the payment of wages (the first payment of the circuit), banks simultaneously debit and credit both firms and workers with purely nominal units of money: the circuit of monetary wages is instantaneous: '[e]very person who is credited by a bank is, in the same moment or circular flow, debited by the same bank for exactly the same amount. Money wages' life span is thus reduced to an instant' (Schmitt 2012, p. 79).

Compliance with double-entry accounting makes it logically compulsory for banks to immediately recover the money they issue to the benefit of workers. There is neither a possible time lag nor a possible numerical discrepancy between the flow and the backflow.

What about the bank deposits wage-earners earn? The payment of wages is a three-dimensional operation. Although wage-earners own deposits with banks, the latter are not net debtors to the former. For every entry made on the liabilities side of a bank's balance sheet there is an equivalent entry on its assets side, since workers' claims on banks are balanced by claims of banks on firms. Thus, workers are creditors of firms through banks. Banks are mere financial intermediaries; they are not the actual source of the credit granted to firms. Money creation must be seen for what it really is: book-keeping entries, that is, debits and credits that banks record in their books in nominal units of account, and that resolve into assets and liabilities denoting financial relations between borrowers and depositors. '[B]anks are only intermediary lenders and borrowers: the initial lenders and the final borrowers are outside banks' (Schmitt 2021, p. 264).[1]

[1]Schmitt was among the very first authors pointing out that money is created within the economy through the actions of commercial banks, rather than being injected into the economy externally by a central bank, a concept that is at the roots of the concept of "endogenous" money. As observed by Rochon (2003), endogenous money challenges the traditional view of money supply as a result of central bank actions alone. Instead, it is seen as the product of credit creation by private banks, who lend based on the demand for loans and the expectations of borrowers and lenders.

Firms owe wage-earners. At the same time, firms have at their disposal the goods produced by workers until they sell them. Production is what makes sense of the instantaneous circuit of money. Banks issue and instantaneously cancel wages in nominal money, but production brings a real content to nominal money. Workers obtain their own product in the payment of monetary wages, instantaneously save it and lend it to firms through banks.

The instantaneous circuit of nominal money takes workers' product into an instantaneous circuit of real money. Workers earn their own product, which they simultaneously deposit with banks and lend to firms. Finally, the income formed by the payment of wages and deposited with banks is spent on purchasing produced goods and services. To that end, income holders (households in general) draw on their deposits. Just as in the payment of wages, final purchases are conveyed through a circular flow of nominal money. Both nominal and real emissions complement each other and overlap in the circular flow of national income.

Before examining Schmitt's positive analysis of national economics further, let us devote the following two sections to his critical analysis of general equilibrium and to the influence exerted by Keynes's analysis on his own macroeconomic approach.

3. Schmitt on the Logical Indeterminacy of Relative Prices

The concept most criticized by Schmitt is that of relative prices as advocated by Walras's general equilibrium analysis (GEA). According to Walras, who considered economics as a 'science which resembles the physico-mathematical sciences in every respect' (Walras 1874 [1954], p. 71), exchanges take place between real goods, and prices are determined through the adjustment of supply and demand by sellers and buyers respectively. Money itself is considered as an asset and its exchange against real goods as a particular kind of relative exchange. Schmitt starts his critical analysis by observing that Walras's aim was that of determining numerical prices through direct exchange, using a mathematical system of independent equations and choosing a good as *numéraire*.

The simplest case analysed by Walras (1874 [1954]) and criticized by Schmitt is that of two agents, A and B, exchanging their goods, *a* and *b*. The system, whose solution is supposed to determine relative prices, is made up of two equations relating to the demand for and the supply of each good. The unknowns are the relative prices of *a* and *b*. In this case, where only two goods are exchanged on the market, only one unknown is left, the relative price of *a* in terms of *b* being necessarily the reverse of the relative price of *b* in terms of *a*. If Walras's system comprised two independent equations, it would be overdetermined, a particular case of indeterminacy, and no solution would be found. This disheartening conclusion is avoided, seemingly, thanks to what is known as 'Walras's law', which states that, at equilibrium, the supply of *a* (*b*) is necessarily equal to the demand for *b* (*a*). Thanks to Walras's law, the system is reduced to one single independent equation and seems apt to determine the relative price of equilibrium that would maximize the utility of A and B.

Schmitt's first critique consists in pointing out that Walras's law is nothing more than a truism that tells us nothing about the relationship between *a* and *b* before exchange takes place. At equilibrium, Walras's identities are tautological. Before equilibrium, in the phase of adjustment where equilibrium should be determined, nothing enables us to establish any kind of relationship between *a* and *b*. The obvious consequence is that, before equilibrium, there are at least two independent equations, one too many for the system to be determined.

Schmitt shows that, in the case of two agents and two goods, the number of independent equations is three: Walras's system is twice over-determined. As a matter of fact, to the two equations that are supposed to determine the equilibrium price of *a* and the equilibrium price of *b* one must add a third equation, equalizing the two equilibrium prices. If we move from the case of two agents and two goods to one considering *n* agents and *n* goods, the analysis remains essentially unaltered, and the conclusion is that the number of independent equations is *n* times greater than the number of variables to be determined.

The idea that, if 6 apples are exchanged for 3 oranges, 1 orange is equal to 2 apples could be retained only if the exchange were capable of transforming oranges into apples, that is, if oranges were apples and apples were oranges. This would indeed be the case if apples and oranges were numbers, in other words, if exchange could make explicit their implicit numerical nature. However, as Schmitt claims, real goods cannot be transformed into pure numbers through their reciprocal exchange, which leaves them physically unaltered.

Schmitt's interest in the neoclassical theory of relative exchanges never faded and in 2012 he published a seminal contribution (Schmitt 2012) in which he showed that GEA is characterized by a mathematical error. This is not an error of mathematics, but of its application to economic analysis. The fact is that relative prices are inversely correlated. Thus, for example, if the relative price of *a* in terms of *b* is doubled, the corresponding relative price of *b* in terms of *a* is immediately divided by two. This means that if *a* is chosen as the *numéraire*-good and if the price of *a* in terms of *b* is made to vary continuously, one price out of two (of *b* in terms of *a*) goes missing and, conversely, that one out of two of the prices of *a* in terms of *b* goes missing when *b* is chosen as the *numéraire*-good (see Schmitt 2012, p. 32). If one price out of two is missing, it is logically impossible to mathematically determine the relative prices of *a* and *b*.

It is important to observe that the logical impossibility to determine relative prices concerns only the neoclassical attempt to determine them through direct exchange. Indeed, relative prices can always be determined starting from absolute prices. Physical products can be made homogeneous and transformed into economic goods only through their absolute exchange, that is, the exchange of each of them against itself. Far from being self-contradictory, this operation is the transformation of physical objects into sums of money. The payment of wages is the absolute exchange transforming production into a creation of money. Economic goods acquire a numerical expression, and it is easy to see that they can immediately be compared, and their relative prices derived from their absolute prices.

4. Schmitt and Keynes

According to Keynes, writing his *General Theory* 'has been for the author a long struggle of escape […] from habitual modes of thought and expression' (Keynes 1936, p. viii). He suspected that his fellow economists would hover 'between a belief that I am quite wrong and a belief that I am saying nothing new' (Keynes 1936, p. v). The second alternative was to dominate. The standard interpretation of Keynesian economics, namely the neoclassical synthesis, is intended to reconcile general equilibrium analysis and Keynes's theory. The latter would be no more than a special case of the former, its originality being to establish a model by which quantities instead of prices are supposed to balance each other (see Snowdon and Vane 2005, p. 71). This is all the more surprising that in the years of preparation of his 1936 book, Keynes asserted that the economy is not the 'real exchange economy' described by the neoclassical school, but 'a monetary economy of production', which he also labelled 'money-wage or entrepreneur economy' (Keynes 1933a, 1933b).

According to GEA, markets are at equilibrium when supply and demand are equal for each productive service and product, and when 'the selling prices of the products [are] equal to the cost of the services employed in making them' (Walras 1926[1959], pp. 253–254). As Walras argued, at that point the theory of production is just an extension of the theory of exchange. In this view, money is but a means to achieve trade in real goods and services, hence the 'real exchange economy' Keynes rejected. Entrepreneurs, for their part, act as neutral intermediaries between the market for productive services and the market for produced goods.

In the monetary economy of production depicted by Keynes (1933a, 1933b), production, hence employment, is not determined by the interplay of supply and demand in correlated markets, but unilaterally by entrepreneurs, who make decisions with reference to the demand they are expecting for their output. In *The General Theory*, the principle of effective demand is in direct relation to this approach. The successive spending and recovering of the money invested in the production process, and the correlated search for profit, are centre stage in the determination of the level of employment.

Schmitt rejects the neoclassical attempt to interpret Keynes's thought by applying the microeconomic approach promoted by GEA and does not miss the macroeconomic relevance of Keynes's analysis. As a matter of fact, Keynes's work had a great impact on the young Schmitt, who repeatedly alleged the fact that he had his first intuitions about the nature of macroeconomics reading Chapter 10 of Keynes's *Treatise on Money* and Chapter 6 of his *General Theory*, where he introduces his fundamental equations.

Keynes argued that the income generated by productive services is always equal to the value of output. This relationship underscores that income serves as both the measure of demand for output and the measure of its value in the product market. Keynes's core equations establish an identity between the value of national product (global supply) and the income required for its purchase, positioning income as the defining factor of macroeconomic demand. According to Schmitt, Keynes's analysis implies a conceptual symmetry between supply and demand, where neither can be considered without reference to the other. Income, both real and monetary, bridges these two concepts. Keynes focused on the demand side, pointing out the concept of effective demand, and the identity he established reflects the simultaneity of output and the income required to purchase it.

As Schmitt pointed out, it is production that, through the intermediation of banks, generates income deposits and makes demand effective. Similarly, supply is effective at the same moment as production, defined by the payment of wages, when physical products and money come together. It is so that effective demand and supply are determined by actual production. Keynes's concept of effective demand aligns with firms' expectations of sales, but it is only after production takes place that income is formed, enabling demand to be financed. Expectations affect firms' production decisions, influencing effective demand, but, according to Schmitt, this demand becomes actual only once income is produced.

In Keynes's framework, income is defined as the value of output, which equals consumption plus investment (Keynes 1936, p. 63). Schmitt had no doubt about the key role identities play in macroeconomics and went on to prove that Keynes's equalities between C + I and C' + I' (that is, between the demand for consumption- and investment-goods and their cost of production) and between S (saving) and I (investment) are identities and not conditions of equilibrium. Schmitt's theory of investment is thus based on such identities.

5. Schmitt's Theory of Investment

Schmitt's economic analysis is known as a quantum macroeconomic approach to money, production, and capital. The idea, already present in his early works and fully elaborated in his 1984 volume (Schmitt 1984c, 2021) on inflation, unemployment, and capital malformations (see the Appendix), is that economic magnitudes are best conceived as the objects of simultaneous creations and destructions, that is, emissions.

As pointed out, bank money is Schmitt's starting point. To consider money as a positive creation of banks would be to accept the metaphysical idea that a human institution can create something positive out of nothing, which is an obvious nonsense. The only logical solution is to think of money as the result of an emission, that is, of an instantaneous creation-destruction. The same applies to production. If its result, namely the product, were to define a new matter or energy, production would pertain to the category of miracles, which is absurd. Metaphysics is avoided if economic production is conceived as an emission. What is created and immediately cleared is neither matter nor energy but a form, and production is the instantaneous event through which matter or energy is given a new form. It is only at the very instant the transformation is completed that production takes place.

The creation-destruction of the economic product is the flow aspect of the emission that defines production: its wave-like aspect. If the physical transformation takes a month to complete, at the end of the process production defines the instantaneous covering of the month in a circular movement to and from. Now, this circular flow is what determines the corpuscular aspect of the emission: the instantaneous emission of the whole month as a quantum of time.

In a succession of logically entangled steps, Schmitt demonstrates first that expenditures are also inscribed in quantum time, each of them being a wave-like movement that defines the emission of the sum spent as a quantum. He then shows that, through the emission of bank money and the payment of wages, the product — a quantum of time — is emitted as a sum of money-wages. It thus follows that production is identified with the (instantaneous) payment through which physical output becomes the real content of a sum of money-income. Schmitt's next step is to show that income, too, is the object of an emission, that is, it is created-cleared even though consumption, which defines the destruction of income through its final expenditure in the products market, does not coincide with production in chronological time. We have here another example of the originality of Schmitt's thought. Thanks to quantum macroeconomic analysis, he can show that the correct understanding of economics requires the use of a logic more advanced than traditional binary logic. Rejecting the principle of the excluded middle, quantum logic enables Schmitt to explain that production and consumption are at the same time chronologically distinct and simultaneous: they are distinct in continuous time, but they are simultaneous in quantum time (for more details see Cencini 2023).

Schmitt's analysis develops further along these lines showing, among other things, that profit is incorporated in the circuit of wages and, more generally, that [*t*]*he theory of emissions is the only one capable of explaining the existence of non-wage incomes* (Schmitt 2021, p. 62, italics in the original).

In coherence with labour being the sole macroeconomic factor of production, Schmitt emphasized the fact that profit results from a division of a pre-existing income instead of being an additional income. All incomes are initially created as wages, and only in a second stage some of them are changed into profit. By selling goods at a price higher than their production cost, a firm can capture part of households' income and finance (retroactively) an equivalent part of its production. Profit therefore follows the opposite direction to that of wages: the latter are formed in the producing services market (the labour market) and spent in the products market, the former is formed in the products market and spent in the producing services market. Through the formation of a profit that is both monetary and real in nature, firms are left with the ownership of a stock of consumption-goods they can exchange for an equivalent amount of instrumental goods (to be also called hereafter investment goods or fixed capital). In other words, workers producing instrumental goods are paid with profit, an income which, deprived of its real content (that is, instrumental goods), gives them a right to purchase the consumption-goods still stocked with the firms.

In his 1984 volume, Schmitt explains the formation of fixed capital through the investment of profit in the production of instrumental goods taking place in a period following that of the production of consumption-goods, in which the initial stock of firms' goods is formed. Derived from the wages paid in period 1, profit is invested in the production of capital-goods in period 2. Hence, workers are paid in an operation that empties their wages of their content in favour of firms. By paying wages out of their profit, firms purchase the capital-goods produced in period 2: they appropriate them since their very production. Strictly speaking, workers alienate their labour to firms together with the product resulting from their working activity. This does not mean that wage-earners do not get anything in the operation; they can indeed spend their wages to buy the goods initially captured in the formation of profit.[2]

According to Schmitt (1984c, 2021), the formation of fixed capital through firms' final expenditure of profit in the labour market is pathological. Indeed, consistently with positive theory, (a) every production should be financed through banks' credit lines, (b) wage-income should be spent in the purchase of consumption-goods, and (c)

[2]The same results were achieved in a manuscript of 1998, to be published in 2025 (Schmitt 1998, 2025), in which he abandons the analysis based on the analytical distinction between periods and replaces it with a more general one in which everything happens in a single period.

monetary profit, associated with a stock of investment goods, should be kept out of the financial system (to wit, out of ordinary bank lending).

A reform of the national payments system along these lines would impede monetary and real phenomena such as inflation and unemployment, to which Schmitt dedicated much of his efforts over the years (see the Appendix).

6. Schmitt's Reform of the System of National Payments

Schmitt (1984c, 2021) proposed a reform of the national payments system that accounts for the logical distinction between money, income, and capital. The plan he advocated is technical in nature, based on a series of banking accounting rules consistent with the macroeconomic laws uncovered by his positive analysis and seeking to favour the sound functioning of the payments system. The reform would have no impact on the behaviour of economic agents: the activity of households, firms, and the public sector would not be affected by it.

Schmitt's reform envisages the establishment of three bank departments: Department I, or Money Department, which tracks monetary emissions; Department II, also referred to as Saving or Financial Department, which accounts for the formation, expenditure, redistribution, and saving of income; Department III, or Fixed Capital Department, which accounts for the formation of macroeconomic saving, that is, the capital resulting from the investment of profit in the production of instrumental goods. The three departments are interrelated in such a way that any entries in one of them are matched by corresponding entries in at least one of the others, reflecting the logical relationship between (i) monetary emissions, (ii) the creation, lending, and expenditure of income, and (iii) the investment of profit. The way in which money, income, and capital are entered in the three bank ledgers is thus in conformity with the macroeconomic laws underlying Schmitt's analysis.

Banks act as both monetary and financial intermediaries. When they act as monetary intermediaries, they create (and clear) money. When they act as financial intermediaries, they transfer income from savers to borrowers. The distinction between the first two departments is necessary to separate the emission of money (to be recorded in the first department of banks' ledgers) from the financial intermediation of pre-existent savings (to be recorded in the second department). 'No creation is an intermediation. The converse is also true: no intermediation is a creation. Indeed, if banks lend a sum of money they have borrowed, they do not create anything' (Schmitt 2021, p. 192).

Thanks to Department I and Department II, commercial banks are constantly aware of the flux and reflux of money in their accounts, as well as the variation in savings of their clients. Department II records the income created by productive activity and its variations resulting from consumption, redistribution, and lending. These two departments are specifically conceived as to serve specific functions. Indeed, the existence of the two departments makes sure that ordinary loans are never financed through monetary over-emission but are always fed by current savings. This impedes the financing of consumption expenditures through the emission of purely nominal units of money, which would lead to a general increase in prices. Furthermore, these two departments guarantee that the purchase of goods and services be financed by the income generated through their production, which is consistent with the identity between sales and purchases advocated by Schmitt (1975).

The major function of Department II is to account for the income generated by productive activities and available for the expenditure on final consumption. Hence, as regards the profit of firms, only distributed profit ends up in the second department, which records the profit accruing to firms' stakeholders in the form of interests or dividends. In contrast, the profit invested by firms in the formation of instrumental capital is to be found in Department III. The relationship between the first and the second departments is crucial to make sure that all productive activities concerning the production of consumption- and investment-goods be financed through the activation of credit lines. It has also the purpose of enabling banks to know at any moment the exact amount of income they can lend to the economy. The relationship between the second and third departments is essential to guarantee that under no circumstances the payment of wages be financed by firms' profit. Thanks to the recording of national saving (macroeconomic investment) in the third department, ordinary loans are never financed drawing from the profit invested by firms. In other words, the third department to be created in banks' bookkeeping must make sure that invested profits are withdrawn from financial circulation (captured by the second department — the only department that to date exists in banks' accounting). 'The problem within the actual structure of domestic payments is that the formation of fixed capital does not lead to the capitalisation of profits' (Cencini 1996, p. 59). As a matter of fact, 'it is not the investment of profit that has to be questioned, but the way its entry is recorded in bank accounting' (Cencini 1995, p. 74).

It must be observed that if the reform of the system of national payments proposed by Schmitt were implemented, workers would still receive their wages, and income would still be distributed in the form of rent, interest, and dividends. Firms would still be free to distribute their profits or to invest them in the production of instrumental goods. Thanks to the new bookkeeping rules advocated by the reform plan, banks would serve as neutral intermediaries between firms and households through the application of accounting practices coherent with the nature of money, income, and capital. By eliminating the mechanism responsible for the pathological over-accumulation of capital, Schmitt's reform would thus contribute to the eradication of involuntary unemployment and inflation (see Appendix).

Let us move on now to the nature of international payments and the shortcomings of the present international monetary regime (see Schmitt 1984a, 1984b).

7. The Present Regime of International Payments

The gold standard, which had been the international payment system of the 19th century, reached its final phase in the early 20th century, as countries began abandoning it. According to Keynes, central banking techniques, which were evolving at that time, played a key role in this shift. Keynes opposed the continuation of the gold standard, advocating for a more scientifically managed currency. By the 1920s, it was clear that US monetary policy, not the gold standard, would determine gold prices. Keynes argued that in the modern world of paper currency, there was no escaping a 'managed' currency, as its value depended on central bank policies.

In addition to the decline of the gold standard, the aftermath of World War I led to debates over war reparations (on this, see Beretta 2013; Carrera 2024; Carrera and Cencini, 2024). The Treaty of Versailles and the London Agreements of 1921 imposed heavy reparations on Germany, which Keynes (1919) criticized in *The Economic Consequences of the Peace*. He identified two key issues: the budgetary problem (Germany's need to raise funds through taxes) and the transfer problem (Germany's need to obtain foreign currency and gold to pay reparations). Keynes argued that, in order to acquire the foreign currency needed to pay war reparations, Germany would have been faced with a devaluation of its national currency, which would have multiplied by two the macoreconomic cost of war reparations.

It was in such a context that new proposals for world monetary reform were developed. Keynes (1942) and Schumacher (1943) were among the first economists to emphasize the importance of multilateral clearing in payment systems (see Beretta 2013; Carrera 2024). The clearing system proposed at that time, which aimed to balance the debts and credits of countries making reciprocal payments, was still incomplete, though it was successfully implemented in national banking systems. Schumacher's proposal suggested the creation of a National Compensation Fund in each country, which would receive domestic currency from importers and pay exporters in domestic currency. An International Clearing Office (ICO) would mediate these transactions. Creditors would hold the balances of debtor countries through the National Compensation Fund and the ICO. However, Schumacher's system had a significant flaw, as it required countries to balance trade credits and debts in the long run, a notion that lacks logical foundations in modern banking practices.

Keynes, in contrast, proposed the creation of a new international currency, which he called bancor, for payments between countries within an international monetary union. The bancor would be used in cross-border transactions, while national currencies would remain in use for domestic payments. The primary objective of the bancor was to provide a system of multilateral clearing, allowing countries to use their earnings from exports to purchase goods from other countries. Keynes envisioned the bancor as a financial instrument, with net exporting countries depositing their bancors in the common institution, which would then lend this amount to deficit countries. While Keynes's system of multilateral clearing was innovative, it faced some challenges, including a failure to clearly distinguish between the national and international monetary circuits and an incomplete departure from gold.

Triffin (1960) and Stamp (1963) later proposed alternative reforms to the international payment system. Triffin suggested creating an international currency issued by the International Monetary Fund (IMF) for clearing international payments, but his plan treated the international currency as a reserve asset, not adequately addressing the real payment of international transactions. Stamp's plan involved issuing IMF certificates to increase the purchasing power of developing countries, but like Triffin's plan it did not fully resolve the real exchange of goods and financial assets without creating additional credit.

Ultimately, the ideas advanced by Keynes, Schumacher, Triffin, and Stamp never materialized. At the Bretton Woods conference in 1944, the White Plan prevailed, giving rise to the current international monetary regime. This regime is a 'non-system', as Williamson (1977, p. 73) defines it cogently. Such a regime provides an 'exorbitant privilege' for those few countries — principally the United States — that issue so-called 'key currencies', as it gives rise to trade deficits 'without tears' (Rueff 1963, p. 322).[3] To be sure, this also creates payment deficits as pointed out by Machlup (1963, p. 256), which testify the existence of an international monetary disorder, few countries having the possibility 'to acquire without paying' (Rueff 1971, p. 23). Such a lack of payment finality at the international level implies a process of 'duplication' (Rueff 1963, p. 322): the banking system of the exporting country enters in its ledgers the mere image (a duplicate) of the bank money paid by the importing country (see Schmitt 1975). This duplication testifies that the importing country does not finally pay its debt, as the exporting country merely receives an acknowledgment of debt (an IOU: 'I owe you') from the 'paying' country. This gives rise to a vicious circle, as suggested by Rueff and Hirsch (1965, p. 2): the US economy, for example, lives beyond its means because exporting countries accept the IOUs issued by the US banking system as final payment for their net sales of real goods, services, or financial assets to US residents. This international monetary regime is highly unstable, and its apparent equilibrium is a 'balance of financial terror', as Summers (2006) noticed explaining the existence of global imbalances that can induce systemic financial crises such as the global financial crisis that burst in 2008 (see Cencini and Rossi 2015; Rossi 2011).

Be that as it may, the present 'non-system' of international payments is a source of inflation for all those countries that accept national acknowledgements of debt (IOUs) 'as if' this was enough to pay net imports finally. As Schmitt (1973, 1975) explains, such a lack of payment finality at the international level is a factor of inflation in exporting countries, because their banking systems enter a duplicate of the bank deposits that the importing economies do not lose when their residents pay for their net imports. Such a system has generated enormous imbalances in the amounts of reserves of foreign currencies held by net exporting and net importing countries. By 2024, for instance, China's foreign exchange reserves (excluding gold) have totalled a value of 3.301.320 millions of US dollars, while Bolivia's have reached a preoccupying historical minimum of 241 million US dollars (Lane and Milesi-Ferretti 2024).

The current international monetary regime considers national currencies as if they were objects of trade, whereas their nature is that of means of payment. Now, logically, an item cannot be both a means and an object of payment within the same transaction. Money is thus denatured by the existing 'non-system' of international payments, making it an object of financial speculation across the foreign-exchange markets that operate at the global level (see Schmitt 1973). As a result, foreign-exchange markets are highly volatile, which is both a cause and a consequence of the speculative transactions occurring daily across these markets. In this regard, Schmitt (1973, 1975) explains that the duplication of US bank deposits (or of the bank deposits of any other key-currency country whose global imports exceed its exports) abroad gives rise to a financial capital that induces international speculation: the 'financial bubble' that has generated the global financial crisis in 2008 originated in such a capital, which has no connection to any production across the world.

[3]In the first quarter of 2024, the shares of major key currencies in allocated reserves were as follows (International Monetary Fund 2025): 57.39 per cent (US dollars), 20.02 per cent (euros), 2.17 per cent (Chinese renminbi), 5.82 per cent (Japanese yen), 4.97 per cent (pounds sterling), 2.27 per cent (Australian dollars), 2.74 per cent (Canadian dollars), 0.17 per cent (Swiss francs), 4.46 per cent (other currencies).

The problem stems from the neoclassical conception of relative exchange and the belief that monetized transactions are relative exchanges between money and real goods, where money itself is a good. The current international monetary regime operates therefore as if international payments pertained to a sophisticated kind of barter. As Schmitt shows, to identify money with a net asset is seriously mistaken, which prevents the correct understanding of cross-border payments and how they should be conveyed between nations. Hence, for example, the lack of a true supranational currency has the unsettling consequence of depriving the system of a standard capable to make currencies homogeneous. Today's international monetary disorder elicits a series of macroeconomic issues like inflation and systemic financial crises, and calls for an appropriately designed monetary–structural reform. We shall analyse it in the next section dedicated to the normative aspect of Schmitt's analysis.

8. Schmitt on Countries' Sovereign Debt

The analysis of the present non-system of international payments led Schmitt to consider the problem of the very formation of countries' external debts and of their remuneration. For more than a decade, he focused his attention on the payment of interest on countries' net foreign debt and showed that the mechanism founded on the use of one or more national currencies as international currency is such that the cost of interest payments is doubled. This is so because, in the absence of a true system of international payments, cross-border interest must be paid in real terms and, additionally, in monetary terms. Unable to obtain, cost free, the vehicular currency necessary to convey their foreign payments, indebted countries whose domestic monies are not key currencies must purchase, at a cost, the currencies of their foreign partners.

In the last fifteen years of his life, Schmitt switched his analysis from the double payment of interest to the duplication of the debt on which this interest must be paid. Since paying twice the amount of interest owed on a debt of x units is equivalent to paying interest once on a debt of $2x$, Schmitt set to work to show that the formation of countries' external debt is itself pathological. Indeed, countries' foreign debt should only mirror the debt incurred by their residents. This is not what happens today, since countries (notably their banking systems) run themselves into a foreign debt, which is in every respect additional to the one incurred by their domestic economies.[4]

The concept of sovereign debt is not the same as the concept of public debt. The latter defines the debt incurred by the State to the purchasers, domestic or foreign, of its financial assets (mainly government bonds), whereas the former is the debt weighing on the country as a whole. Any State that needs to finance its expenditures through a sale of financial claims runs a public debt. Yet, this debt concerns a resident (the State) and not the country itself, considered as the macroeconomic set of all its residents. The external debt incurred by any resident is the logical consequence of its sales of financial claims abroad. On the contrary, any sovereign debt incurred by a country is the result of a pathology that multiplies by two the charge of a country's net external purchases.

[4]The expression 'sovereign debt' acquires a special meaning here. It refers to a debt that has no logical reason to exist, representing the unjustified indebtedness of the country considered as a whole. On the other hand, the expression 'external debt' refers here to the debt incurred by the residents of any given country because of their net foreign purchases.

The first charge of a country's net purchases abroad weighs on the country's importers, who pay for their purchases in domestic currency. The second charge weighs on the country itself, whose banking system must sacrifice an equivalent amount of foreign currency to convey the payment to foreign exporters. In a long paper that he finished writing just a few days before his death, Schmitt provides various proofs of the pathological formation of countries' sovereign debt (Schmitt 2014). Let us summarize two of them. The first refers to the need, within the present non-system, to pay for a country's net imports both in real and in monetary terms. The payment in real terms is a necessary consequence of the identity between any country's global imports and exports. According to this identity, any country's net commercial purchases (sales) are necessarily balanced by an equivalent sale (purchase) of financial assets. It is enough to observe that any financial claim has a (future) product as its real object to realize that the first debt incurred by a country whose residents are net commercial importers defines the real payment of its residents' net purchases. If the system of international payments were sound, this would be the only payment weighing on the country. If this were the case, no sovereign debt would form in the system.

Today, another payment adds up. This is so because foreign exporters must be paid in their own domestic currency. The first loan obtained abroad being necessary to balance the purchase of foreign goods with an equivalent sale of (future) domestic goods, a second foreign loan is necessary to provide the country with the foreign currency required to pay, monetarily, its net purchases. This second loan is what explains the pathological formation of countries' sovereign debt.

Schmitt's second proof is straightforward. It rests on Newton's third law of motion, which states that for every action there is an equal and opposite reaction (Newton 1687 [1999]). Applied to international transactions, this law establishes that the action that leads to an increase in imports reduces the amount of foreign currency obtained from the country's exports (reaction). If we start from equilibrium and assume that country A's imports increase by x, we observe that its exports decrease by x so that the difference between imports and exports is equal to $2x$. To pay for its net imports of x units, the country must therefore borrow $2x$ abroad. The first sum of x units is justified, the second is unjustified and corresponds to the country's sovereign debt.

Let us now consider the normative aspect of Schmitt's analysis, with particular attention to his two proposals for a reform of the international payments system.

9. Toward a New World Monetary Order

9.1. A Multi-country Reform of the International Payments System

More than half a century ago, Schmitt (1973) laid the foundation for a world monetary reform, which could be implemented by different groups of countries or by them all at once. Compared to the major reform proposals of the past (Keynes 1942; Rueff 1963; Schumacher 1943; Triffin 1960, 1963), Schmitt's plan introduced novel elements that can be better understood by referring to the working of national monetary systems.

In any country, commercial banks issue money in the form of their own acknowledgment of debt (IOU). Under no circumstances do commercial banks directly settle payments among themselves using their IOUs. In fact, the central bank acts as the common intermediary of commercial banks. Through a system of settlement and clearing, commercial bank monies are transformed into central bank money, ensuring monetary homogeneity within the national territory. The intermediation of the central bank guarantees that payments among commercial banks are effective, while the national currency becomes the vehicle through which income is transferred from one economic agent to another, ultimately to be spent on the final purchase of real products. Money itself is never the object of payment but rather the means through which payments are carried out: its use is circular and instantaneous. National payments are real not because money is their final object, but because money immediately gives way to bank deposits, whose real content is the part of national output with which money was initially associated.

Arguably, an attempt to replicate the functioning of national banking systems on an international scale has been made with TARGET2, the payment system supported by the European Central Bank (ECB). However, this system does not guarantee the effective settlement on a multilateral basis among the banking systems of the countries adhering to it. In fact, payments among residents of member countries take place almost entirely directly, without the monetary and financial intermediations of the ECB. Things are not any different in other world areas. Never has a system of settlement and clearing been adopted as a bridge between national monetary spaces. The international monetary order proposed by Schmitt instead moves in that direction.

As previously done by Keynes (1942) in his proposal for the creation of an International Clearing Union, Schmitt argued that national currencies must be used as vehicular means only within their own national banking systems. Both Keynes and Schmitt were convinced that a proper international payment system requires the creation of a supranational bank charged to issue a supranational currency, which would serve as a bridge between national banking systems and national currencies.[5]

Residents of any country would pay and would be paid in their own national currency. Yet, cross-border payments would be conveyed through the vehicular use of the supranational currency. Central banks, together with the supranational institution, would carry out transnational payments through the conversion of national currencies into the supranational money. Real payment would be guaranteed by a multilateral system of real-time gross settlement (RTGS) in which the supranational currency would be a mere monetary instrument and never the object of payments. The supranational bank would use the international currency in a simultaneous flux-reflux so that real payments between countries would be reciprocal, any country's net commercial imports being balanced by equivalent exports of financial claims.

> Only the equality of sales and purchases of each country in each period gives assurance that the surplus countries are actually paid by the deficit countries. The supranational currency thus establishes the solidarity of countries, by bringing together their opposing needs in the financial market. (Schmitt 1977, p. 154, our translation)

In addition to serving as a monetary intermediary, the supranational bank could also act as a financial institution to the benefit of those countries that would find it difficult to sell

[5]This idea has been revived in the aftermath of the global financial crisis of 2008, for example by the then Governor of the Chinese central bank (see Xiaochuan 2009), and more recently by the governments of Argentina and Brazil (see Government of Argentina 2023), among others.

their own financial claims in the foreign-exchange market. Exchange rate stability between national currencies and the supranational money would be the rule, since national currencies would no longer be traded in the financial market as if they were net assets. In this connection, let us point out that the global solution proposed by Schmitt could also be implemented by a group of countries instead than by all of them. Such a 'regional', multi-country reform would benefit member countries both when settling payments among themselves and when paying non-member countries. Transnational payments between member countries would be carried out in a new, common currency and so would the external payments of member countries, which would thus avoid the shortcomings deriving from the lack of a true system of international payments (see Carrera and Cencini, 2024, for analytical elaboration on the creation of a Clearing International Monetary Area, CIMA).

9.2. A One-Country Reform of the System of International Payments

Aware that the implementation of a worldwide reform of international payments is not for tomorrow, Schmitt worked out a plan that would enable each single country to protect itself against the formation of a sovereign debt, that is, against pathological over-indebtedness, while paying foreign creditors their full due. According to Schmitt's plan, every single cross-border payment would have to be conveyed through the intermediation of a national *Bureau* managed by the country's central bank. Each importer's payment would be carried out by their commercial bank, in domestic currency, to the benefit of the *Bureau*. On the other hand, country's exporters would be paid, again in domestic currency, by the *Bureau*. It immediately appears that, in case of a country's net imports, the *Bureau* would benefit from a gain equal to the difference between the amount paid by importers and that paid to exporters. Yet, something more is necessary to avoid that this gain be appropriated by the rest of the world (country R). In the absence of a mechanism preventing the formation of the country's sovereign debt, the gain of the *Bureau* would be lost to R, which would obtain it as the real content or object of the external debt incurred by the country (A) because of its net foreign purchases.

The mechanism proposed by Schmitt (2014) is a counter-loan granted by country A's *Bureau* to country R. The loan of R to A is what enables country A to obtain the foreign currency needed to pay for its net purchases. The counter-loan of A to R is what balances A's net purchases from R with equivalent sales of A's future output. Through this counter-loan, country A obtains the double result of paying in real terms its foreign purchases and avoiding the formation of a sovereign debt. If a true system of international payments were adopted by a group of countries, none of them would have to implement the mechanism of the counter-loan when paying another member of the group. A RTGS system would guarantee the real payment of cross-border transactions through the vehicular use of a purely numerical standard. In the absence of such a system, every country willing to protect itself against the pathologies arising from the formation of a sovereign debt would need to create a mechanism that would immediately counter-balance the loan granted by R to A.

An example of the way country A would pay for its net purchases illustrates the practical working of Schmitt's one-country reform.

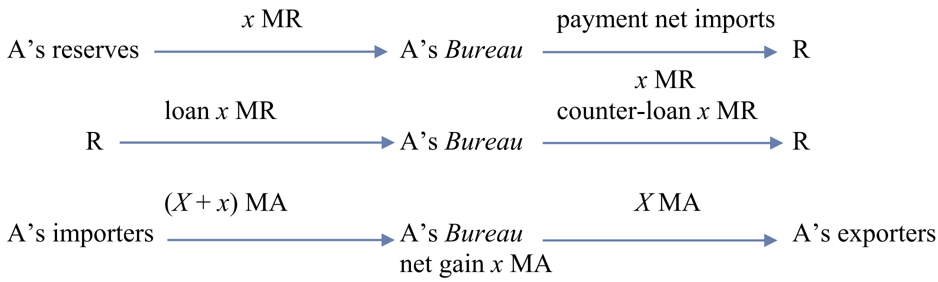

Figure 1. Payments of A's imports in period p_0.

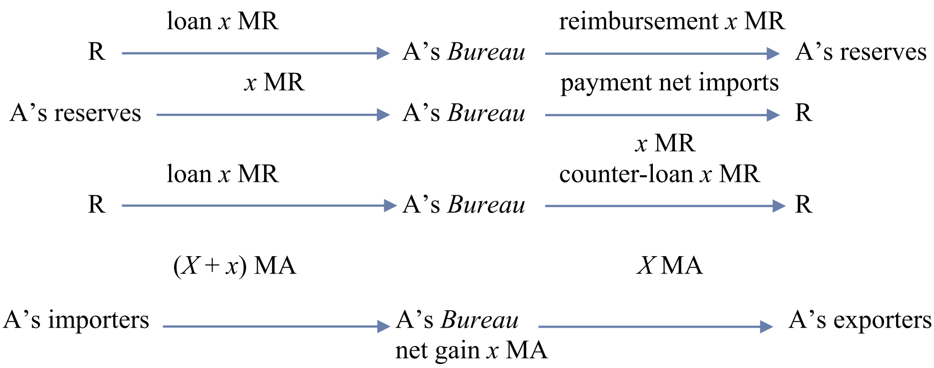

Figure 2. Payment of A's imports in every period following p_0.

Suppose that country A's global imports from R are greater than its exports, that the difference is equal to x units of A's domestic currency, MA, and that the rate of exchange between A's and R's currencies is of 1 unit of MA for 1 unit of money R, MR. In period p_0, when A's trade deficit forms for the first time, payments would be carried out as Figure 1 shows.

At the end of period p_0, the *Bureau* of country A has a debt of 1 MR *vis-à-vis* A's official reserves balanced by a gain of x MA. But it is from the following period, p_1, that the system would produce its effects fully, as Figure 2 shows.

At the end of p_1 and of any following period, and assuming that in each period country A's net imports are equal to x units, the debt of A's *Bureau* to the official reserves would remain unaltered, while a new net gain of x MA would enable the country to increase its national production period after period. Asymptotically, in the long run, the debt of A's reserves would tend to zero. Country R would be paid fully for its net exports, because it would obtain an equivalent amount of A's future exports through A's *Bureau* counter-loan. At the same time, country A would not incur any sovereign debt and would not lose even a cent of its national income, the extra payment of its importers being gained by A's *Bureau*.

10. Conclusion

Bernard Schmitt's death in 2014 ended his fundamental contributions to macroeconomic analysis. However, his ideas are still alive and will continue being the source for the development of his theory in the years to come. This paper has been written with the purpose to introduce the reader to Schmitt's quantum macroeconomics and to explain the profound change it entails compared to mainstream economics. The best way to honour Bernard Schmitt ten years after his death is to show that his analysis provides the right answers to the problems facing our economic systems, both nationally and internationally. This paper is just a step in this direction. Each of the topics discussed here needs to be investigated further, along the lines followed by Schmitt in his more than sixty years long research programme. His legacy consists of thousands of pages of published and unpublished manuscripts waiting to be thoroughly examined and

Table 1. A comparison of research topics for Schmitt's scholars and the mainstream.

Concept	Schmitt's theory	Mainstream theory
National economics		
Money	Unit of account	Asset
	Numerical item	Tangible item
	Vehicular means of payment	Object of payments
Monetary creation	Endogenous (by commercial banks)	Exogenous (by the central bank/the State)
Monetary income	Monetary form of output	A positive stock of money
Factors of production	Labour	Several
Value theory	Objective	Subjective
Production	At the origin of economic value	Supply of physical goods
Wages	Nominal and real	Remuneration of labour
	Measure of value	One of many incomes
Prices	Absolute	Relative
Consumption	Final expenditure of income	Generating further income
	Reimbursement of loans to production	
Profit (macroeconomic)	Captured from wages	Income of the capitalist (microeconomic interest)
	Partly invested, partly distributed	
	Positive: monetary and real	Null at macroeconomic level
Investment or fixed capital	Profit investment	No theory of distribution
		Mechanical modes leading to the existing of consumption- and capital-goods
Inflation	Formation of "empty" money (monetary creation not backed by produced output)	Defined by price increase
		Many types (demand-pull, cost-push, etc.)
	It leads to a price increase *à la* Ricardo	
Bank departments envisioned (national systems)	Three: money-, financial- or income-, fixed capital- departments	Financial or income department
International economics		
Money	A vehicular means	An asset
Call for international money	Yes	No
Call for international real-time gross settlement (RTGS)	Yes	No
Inflation	Due to currency duplication	Imported
	Linked to the formation of euro- or xeno-currencies	
Financial bubble	Linked to the investment of euro- or xeno-currencies	Due to agents' behaviour
		Result of supply-demand adjustment
Exchange rate fluctuation	Linked to the investment of euro- or xeno-currencies	Several reasons (trading, geopolitical, expectations, etc.)
Use of reserve currencies	No	Yes
Use of gold	No	Yes / In certain circumstances
Sovereign debt	Debt of the country as a whole. Fully unjustified	Fully justified
International exchanges	Real goods against real goods (financial assets included)	Commercial products against monetary transfers

Source: authors' elaboration.

fully understood. Our contribution will achieve its objective if it can motivate the readers to follow Schmitt's quest for a unified macroeconomic analysis that can explain the current economic disorders and provide a remedy that ensures the evolution from capitalism to post-capitalism.

To date, current developments by Schmitt's scholars are centred on the analysis of balance-of-payment data and countries' gross external debt positions, with the aim of empirically demonstrating the existence of a sovereign debt in net-importing countries. Further, for these scholars it is essential to continue refining reform plans for national and international payment systems, to make them feasible with the existing technology and its most recent developments as regards the payment system infrastructures — like instant payment protocols and (central bank) digital currencies (see Rossi 2025).

Table 1 lists the major research topics of such a fully-fledged theory of national and international economics, compared to mainstream or neoclassical theory and may serve as a beacon for future research avenues.

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Appendix

This Appendix provides an overview of Schmitt's perspectives on inflation and unemployment as the consequences of capital accumulation and amortization.

A1. The Traditional Theory of Inflation

The traditional theory of inflation simply defines the latter as an increase in the general price level, measured across the market for produced goods and services with different price indices, such as the consumer price index or the personal consumption expenditure price index. Such an approach to inflation is wrong, as there are several factors that can increase the general price level, apart from the loss in money's purchasing power. To be sure, if any number of firms increase their selling prices to earn a higher profit, such a decision can increase the price level across the market for produced goods and services, even though there is no inflation properly defined, because money does not lose its own purchasing power — only functional income distribution is affected, with an increase in the profit share and a corresponding reduction of the wage share. Something similar occurs when the general government sector increases the so-called value added tax (VAT): the price level across the product market increases, without there being any inflation, because money does not lose its purchasing power — only national income is differently distributed, so that the public sector earns more fiscal revenue, which indeed is an income transferred from private agents to the fiscal authority. In all these cases, the relationship between money and output (hence also the relationship between money and income) is unaffected because total demand and total supply are identically equivalent — as Keynes's (1930 [1971]) "fundamental equation" shows (see Schmitt 1984c).

The traditional theory of inflation also imagines that there may exist a relationship between the prices of imported goods and inflation — the so-called phenomenon of "imported inflation", which mainstream economists have been pointing out in the past and have relaunched in the aftermath of the war in Ukraine that started on 24 February 2022. In fact, Schmitt (1984c, 2021) explains that there can be no "imported inflation" logically, because inflation results from a loss in the purchasing power of each unit of domestic currency when there are emissions of "empty money". As quantum macroeconomics shows, inflation occurs when the relationship between the number of money units issued in a country and its gross domestic product (that is, produced output) is affected by the emission of "empty money" through the double-entry bookkeeping system of the banking sector. Hence, no increase in the prices of imported goods can induce inflation, as this increase does not affect the money–output relationship established by the payment of wages defining both national income and national output.

The traditional theory of inflation is problematic also as regards monetary policy goals. If the latter aim at price stability across the market for produced goods and services but with such a simplistic understanding of inflation (confusing it with price increases that, as noticed, can stem from a number of factors that have nothing to do with a loss in the purchasing power of money), then monetary policy decisions and implementations can miss the point, as they focus on the symptom (price increases) rather than on its causes eventually. This is the so-called "Gibson paradox" (see, for instance, Levrero 2023): if the central bank increases the policy rate of interest to curb inflationary pressures that in fact are just an increase in the price level resulting from cost-push (or greedy) reasons, its decision will push up the price level further, since firms will have to pay higher interest rates when they ask banks to open new credit lines to them, so that these higher interest payments will be transferred to these firms' selling prices, inducing thereby a new increase in the price level across the market for produced goods and services.

Schmitt (1960) criticized this approach, pointing out its wrong conception of money's nature: the traditional theory of inflation assumes that money is created by the central bank exogenously and circulates through the banking system; the so-called "money multiplier" then provides the total money supply (see, for instance, Friedman 1959). In fact, money is an endogenous magnitude that is issued every time a bank grants a credit line to any kind of economic agents (see Schmitt 1972).

A2. Schmitt's Analysis of Inflation

Inflation is a loss in money's purchasing power, which induces an upward pressure on prices. This essential definition of such a macroeconomic disorder has been pointed out since the early writings of Schmitt (1972, p. 156). Quantum macroeconomics provides a monetary–structural explanation of inflationary pressures, as they result from the lack of distinction in banks' double-entry bookkeeping system between the emission of money, the provision of credit lines for different transactions — including financial activities that have no link with the production of national income — and the investment of profit to produce fixed capital goods.

In this regard, Schmitt (1984c) distinguishes between "benign" and "malign" inflation. The former occurs when banks grant some credit lines *ex-nihilo* for 'non-GDP-based transactions' (Werner 2011, p. 29) across financial markets: when these credit lines are reimbursed, banks clear a corresponding amount of deposits, thereby eliminating the "empty money" originating them. By contrast, "malign inflation" occurs when firms pay their wage bill with some pre-existent profit, thereby purchasing their workers' activity (hence, also its result). In such a case, the emission of wages 'is *empty* as far as it identifies itself with the expenditure of profit' (Schmitt 2021, p. 123, original emphasis). This gives rise to an amount of bank deposits to which no new saleable output corresponds, as this output has already been purchased by firms' expenditure of their profit in the labour market. 'Although total supply and total demand are identically equivalent as measured by the wage bill, total demand implies a sum of empty money that elicits a pathological disequilibrium between money and current output' (Rossi 2001, p. 151). Carried out by de-personalized firms, the investment of profit in the labour market removes indeed the newly produced output by wage-earners from saleable output since firms purchase these new investment-goods in such a market at the very instant of time when their wage bill is paid by any bank transfer. 'This collecting leaves intact *monetary* wages; it only reduces the product within monetary wages' (Schmitt 2021, p. 124, original emphasis). Hence, a pathological disequilibrium occurs between money and output, somewhat analogously to what Keynes (1930 [1971], Chapters 10–11) called "profit inflation" in his own analysis of the investment of profit in the production process (Keynes 1930 [1971], pp. 135–138): in this regard, inflation does not arise simply because 'too much money chases too few goods' (Friedman 1960), but because a share of newly-produced output is not for sale to income holders.

Now, since investment-goods become the definitive property of *a 'non-person', [that is] the disembodied set of the country's firms*' (Schmitt 2021, p. 124, italics in the original), the emission of "empty money" leads to a permanent monetary disequilibrium once it is referred to the amortization of the fixed capital resulting from the investment of profit. Were investment-goods, that is, the goods produced via the investment of profit, entirely owned by "embodied" firms (namely, their shareholders), our economic system would not suffer the disorder elicited by the present recording of capital accumulation in banks' bookkeeping. Indeed, this macroeconomic disorder is not problematic as far as the formation of firms' profit implies an identically equivalent stock of unsold goods: when this profit is invested in the production of new investment-goods, the resulting sum of "empty money" can be spent by wage-earners to purchase those goods that correspond to the formation of firms' profit. As Schmitt (2021, p. 94) pointed out in this regard, the investment of profit amounts to the conversion of a stock of wage-goods into a stock of capital-goods. In Schmitt's own words, '[t]he wages issued in $p+$ do not contain the capital-goods produced there. However, wage-earners paid in $p+$ grab the wage-goods saved on the product of p, when profit is formed' (Schmitt 2021, p. 123). Now, since capital-goods must be amortized regularly, to preserve their potential output, the production of amortization-goods is problematic, to date, for the relationship between money and output, as it gives rise to a permanent inflationary gap.

Schmitt (1984c) pointed out that the amortization of fixed capital generates a sum of "empty money" that increases total demand but not total supply. To be sure, while the investment of firms' profit gives rise to an "empty money" that can be "filled" by a pre-existent stock of wage-goods — thereby preserving the money–output relationship — fixed capital amortization provides a sum of "empty money" to which no saleable output corresponds irremediably.

> It is fitting to say *irremediably*, because, unlike the empty emission defined by net investment, the empty emission induced by the production of amortization-goods results in a money the emptiness of which is not compensated at all; this time, no savings of wage-goods are waiting to fill in the emptiness. (Schmitt 2021, p. 136, original emphasis)

A3. Schmitt's Analysis of Unemployment

According to Schmitt, it is only at the stage of capital amortization that the structural faults in the economic system become apparent and involuntary unemployment settles in. The amortization of fixed capital implies a sort of exchange between households and firms: consumers pay the replacement of capital-goods by gross profit but benefit from an enhanced real content of their income while firms lose the obsolete fixed capital but obtain its replacement.

Fixed capital undergoes wear and tear requiring periodical renewal; a gross investment through a gross profit expenditure in the producing services market is needed to produce replacement equipment. Once firms invest their profit in the production of amortization-goods, wage-earners alienate their product, replacement-goods, to (depersonalized) firms, which prevents any direct exchange between households and firms. As firms cannot be both the payer and beneficiary of the payment in the producing services market, direct amortization is not possible, which is why, in today's capitalism, amortization is achieved through the pathological investment of profit. Any profit is derived from wages; yet, in capitalism, its expenditure in the labour market leads to the direct appropriation of workers' output by firms. This occurs both when profit is invested in the production of capital-goods and when it is invested in the production of amortization-goods. Whether the profit invested is net or gross, the result is not substantially different: households are dispossessed of the ownership of fixed capital-goods as well as of the ownership of the replacement-goods required to maintain unaltered the value of accumulated fixed capital.

In his 1984 book, Schmitt shows that the production of amortization-goods gives rise to "empty wages" (and, thus, to inflation) whose expenditure is the source of an additional profit, the investment of which defines a capital over-accumulation. Amortization is therefore a dual phenomenon involving both a production of replacement-goods and a production of new investment-goods. Schmitt's 1998 analysis confirms that of 1984: amortization is a twofold process implying the production of replacement-goods in sector 3 and an increased profit, formed in sector 1 (wage-goods) and spent in sector 2 (investment-goods). However, capital accumulation and over-accumulation cannot go on forever because of the limited amount of profit that can be captured in the economy: profit is derived from wages and the circuit of profit is inserted into the circuit of wages. As Schmitt shows, profit cannot exceed the amount of wages, and the production of amortization-goods (sector 3) cannot exceed one-third of the overall activity of workers. With the passing of time, overaccumulation will therefore become unprofitable, and firms will have to look for an alternative route to increase their profit through financial markets.

What matters here is the difference between the macroeconomic rate of profit, that is, the ratio between profit and accumulated capital, and the market rate of interest. If this difference is positive, firms will invest their profit, included that derived from amortization, in the production of new capital-goods. Yet, when the increase in profit approaches its limit, the rate of profit decreases until it falls below the market rate of interest. When this happens, firms are forced to reduce their productive investment of profit and "park" it instead in the financial market. This "financial" route is the cause of unemployment. Once profits are thrown into financial markets, production is partly abandoned, hence the unemployment of the corresponding workforce.

For a time, firms have a choice: they can either invest their inflationary profit in the production of consumption-goods or "park" it in the financial market. In the first case, however, the increased production of consumption-goods would be the cause of a deflationary process, since the increase in supply would not be matched by an equivalent increase in income. In the present, pathological system, wages paid out of profit are "empty" and cannot feed any increase in global demand. After a while, the lack of global demand characterizing deflation would therefore force firms to reduce their production of consumption-goods and invest their profit financially.

Schmitt's explanation of 1998 follows the same path. The crucial argument remains that of the increasing accumulation of capital and the logical limit imposed in capitalism to the increase of profit. When this limit is reached, production is equally distributed among the three sectors and workers are equally employed in each of them. When the rate of profit falls below the market rate of interest, the production of amortization-goods starts to be incorporated into sector 2. Instead of producing investment-goods, part of the workers employed in sector 2 starts producing replacement-goods. Activity in sector 3 is consequently reduced until the production of amortization-goods is entirely absorbed by sector 2. The reduction in employment has thus a logical limit: it stops when one third of the working population is on the dole.